



Hands-on Lab : Download & install Anaconda

Time efforts: 15 minutes

Objectives of exercise

- Download & install Anaconda
- Create Anaconda Environment for R and Python
- Install and run Jupyter Notebook

Overview of Anaconda

There are several cloud-based data science tools that can make team collaboration more accessible. At times it's useful to work directly on your desktop.

Anaconda Distribution is an Open Source distribution of Python and R languages. It comes with a repository of a large number of packages for data science and machine learning, with the most popular and commonly used ones pre-installed. It includes Anaconda Navigator, a graphical interface (GUI) that contains several tools, and IDEs such as Jupyter Notebooks and R Studio. It has binaries for major platforms, including Windows, Linux, and macOS. This lab includes instructions for downloading and installing Anaconda on Windows.

Exercise 1: Download & Install Anaconda Distribution

Step 1: Use the below link to download the Anaconda distribution:

Link for Download Anaconda Distribution: <https://www.anaconda.com/products/distribution>



Products ▾

Pricing

Solutions ▾

Resources ▾

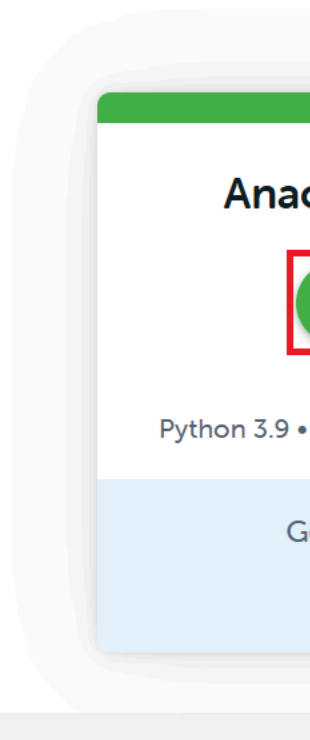
Partners ▾

Blog

Individual Edition is now

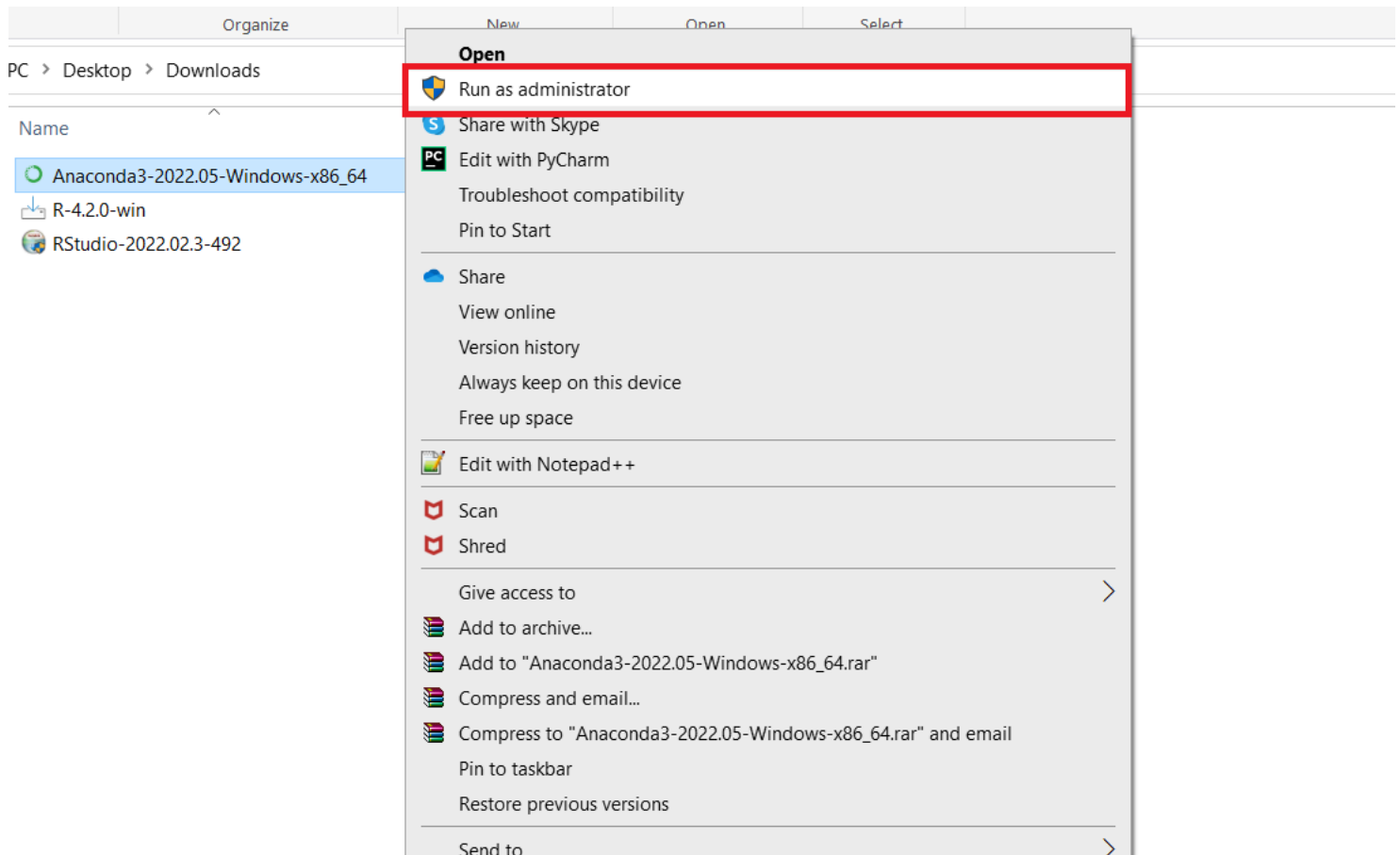
ANACONDA DISTRIBUTION

The world's most popular open-source Python distribution platform



*Note: Depending on your **Operating system**, it would show the download link specific to your OS. Click the **Download** button to download it to your local machine.*

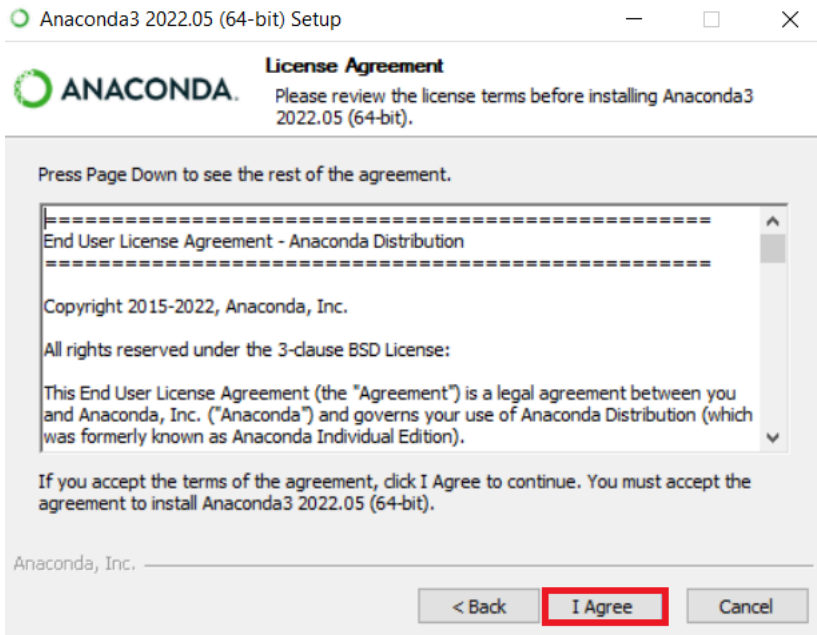
Step 2: Once the download completes, right-click the downloaded file and run it as **Administrator**.



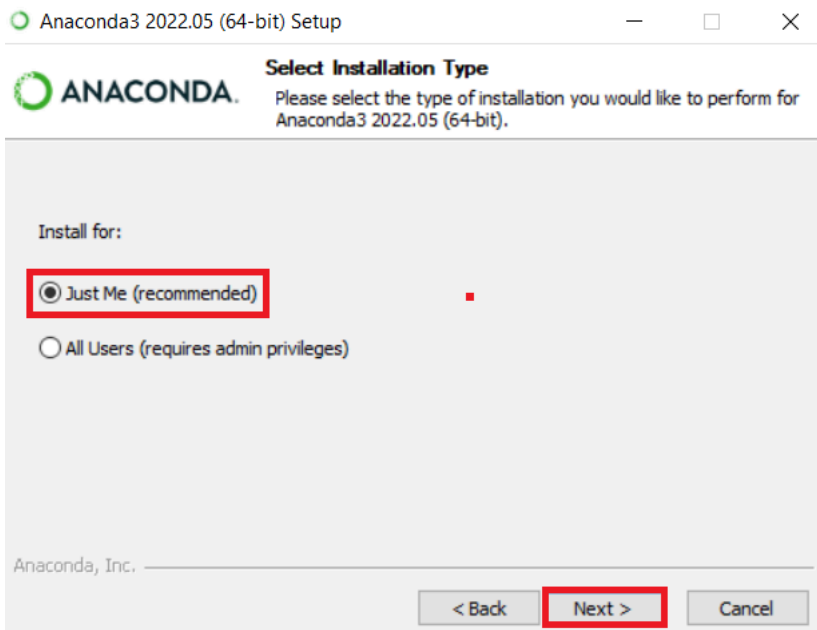
Step 3: At the beginning of the welcome window, you need to click **Next** to confirm the installation.



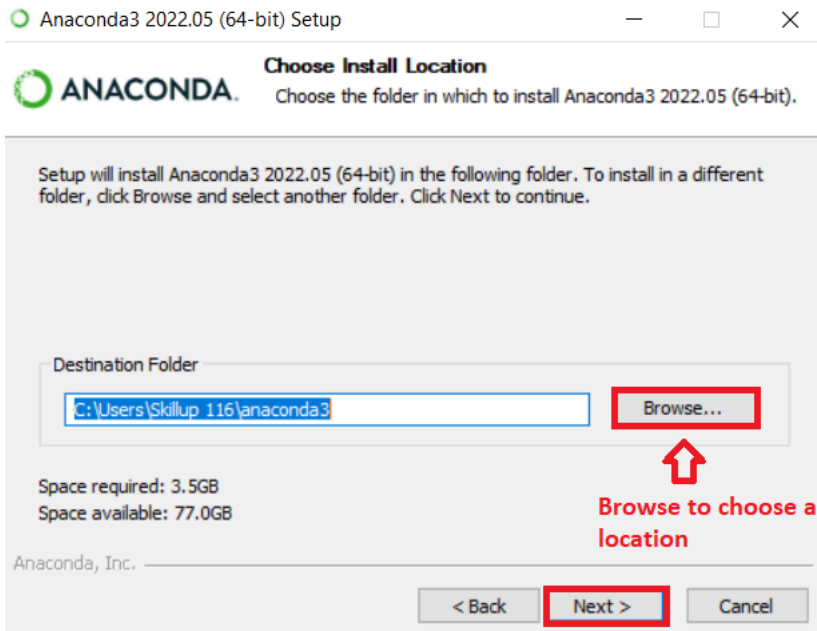
Step 4: Agree to the license.



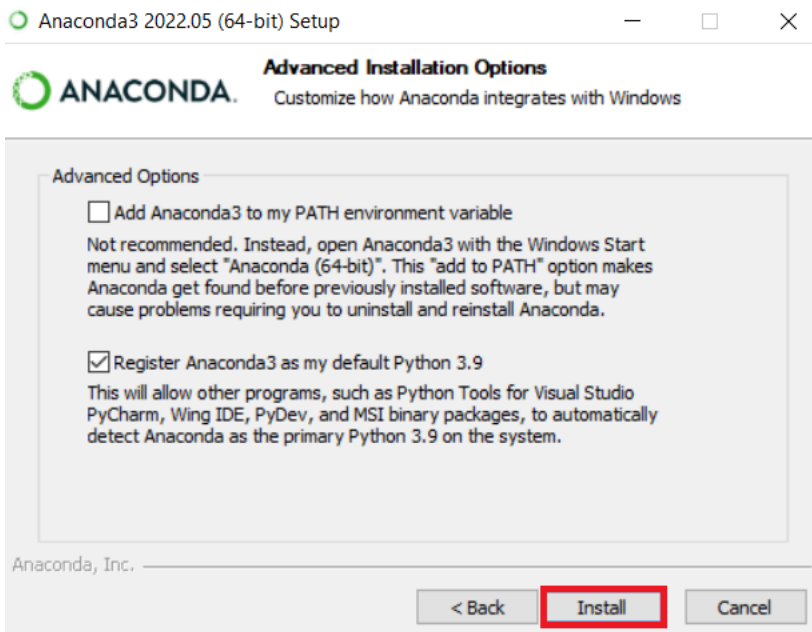
Step 5: In the installation window, select **Just me**, and click **Next**.



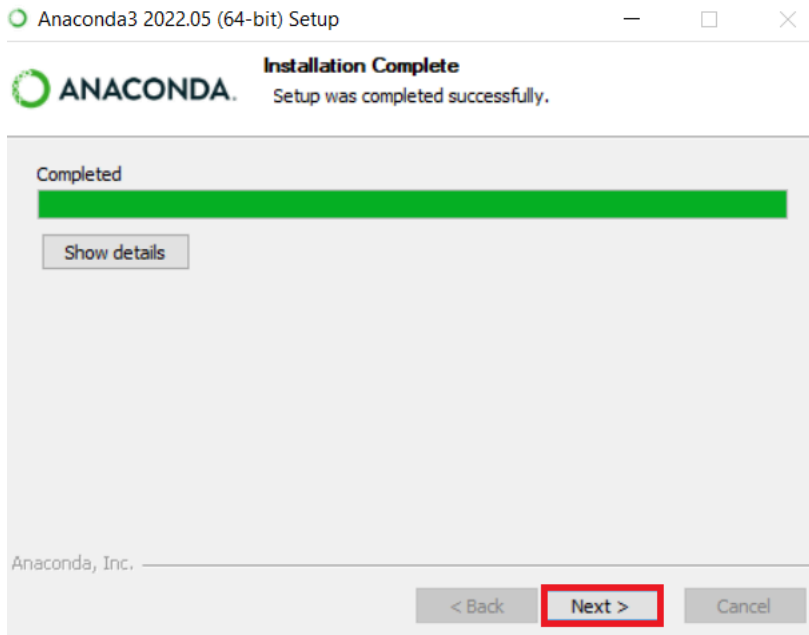
Step 6: Select the folder where you would like to **Install Anaconda**, or retain the **Default** installation location and click **Next**.



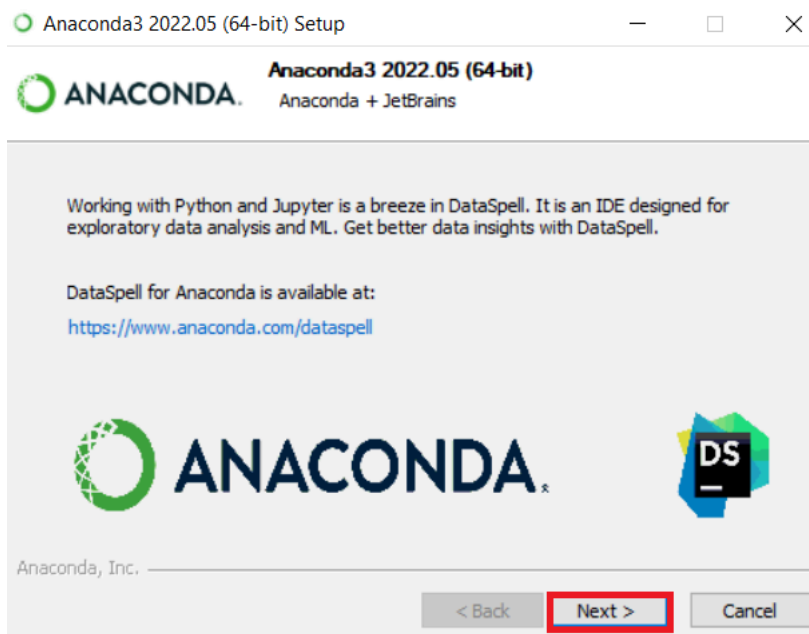
Step 7: In the **Advanced Installation Options** window, select **Register Anaconda3** as the default Python 3.9 option, and click **Install**.



Step 8: You need to wait for the installation to complete. Once installation completes, click **Next**.



Step 9: Click Next.



Step 10: Click **Finish** to complete the installation of the Anaconda distribution.



Completing Anaconda3 2022.05 (64-bit) Setup

Thank you for installing Anaconda Distribution.

Here are some helpful tips and resources to get you started.
We recommend you bookmark these links so you can refer
back to them later.

☒ Anaconda Distribution Tutorial

☒ Getting Started with Anaconda

< Back

Finish

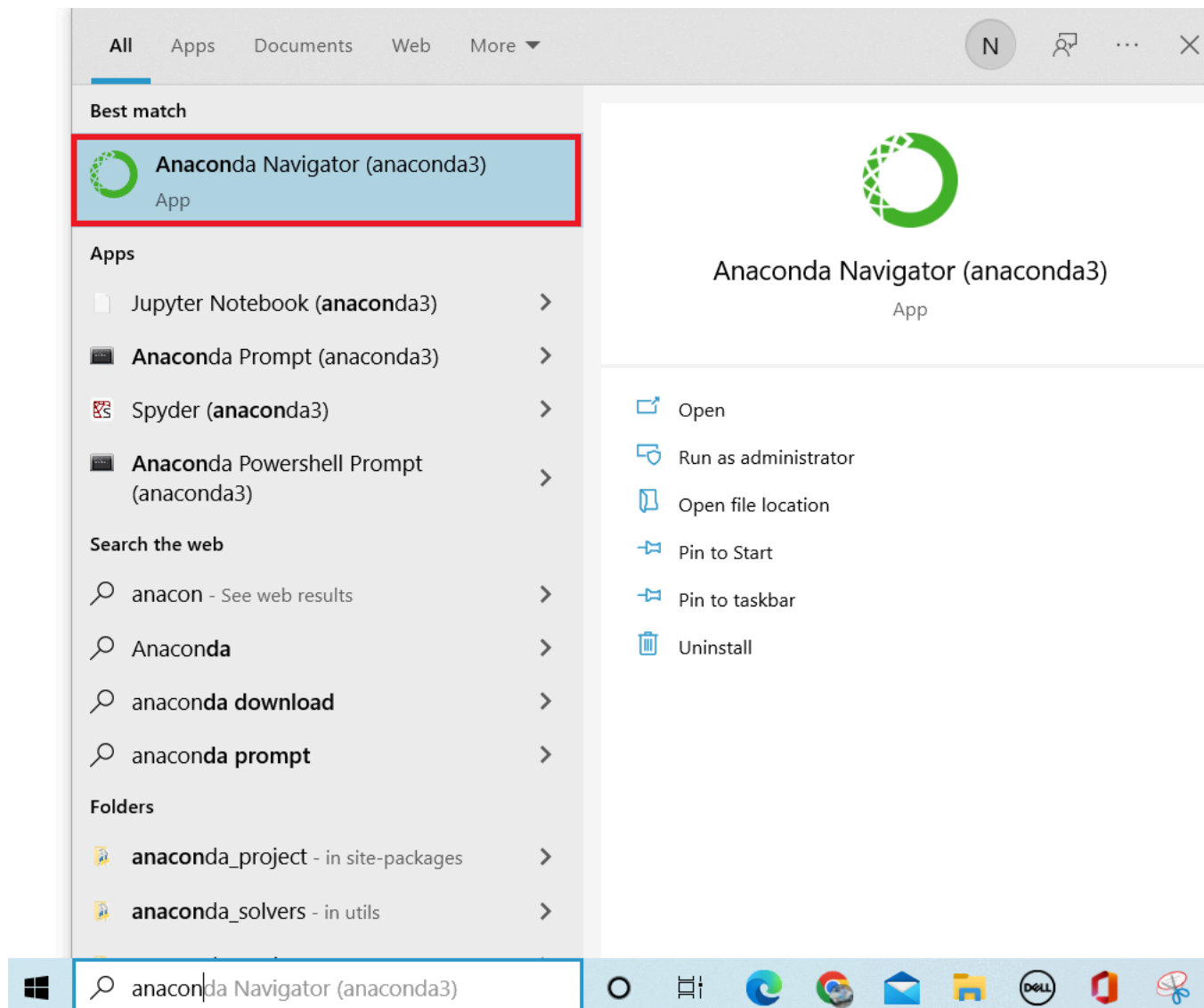
Cancel

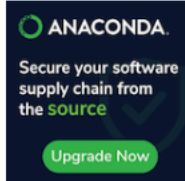
Exercise 2: Create Anaconda Environment

Anaconda environment is a directory containing a specific collection of conda packages you have installed. For example, you may have one environment with NumPy 1.7 and its dependencies and another environment with NumPy 1.6 for legacy testing.

Ref: <https://conda.io/projects/conda/en/latest/user-guide/concepts/environments.html>

Step 1: Open the **Anaconda Navigator** from the Windows Start menu.



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Applications on

base (root)

Channels



CMD.exe Prompt

0.1.1

Run a cmd.exe terminal with your current
environment from Navigator activated

Launch



Datalore

Online Data Analysis Tool with smart
coding assistance by JetBrains. Edit and run
your Python notebooks in the cloud and
share them with your team.

Launch



IBM Watson Studio Cloud

IBM Watson Studio Cloud provides you the
tools to analyze and visualize data, to
cleanse and shape data, to create and train
machine learning models. Prepare data and
build models, using open source data
science tools or visual modeling.

Launch

An extensible e
and reproducibl
Jupyter Not

PyCharm Professional

2021.1.3

A Full-Fledged IDE by JetBrains for both
Scientific and Web Python development.
Supports HTML, JS, and SQL.

Launch



Qt Console

5.3.0

PyQt GUI that supports inline figures,
proper multiline editing with syntax
highlighting, graphical calltips, and more.

Launch



Spyder

5.1.5

Scientific PYthon Development
EnviRnment. Powerful Python IDE with
advanced editing, interactive testing,
debugging and introspection features

Launch

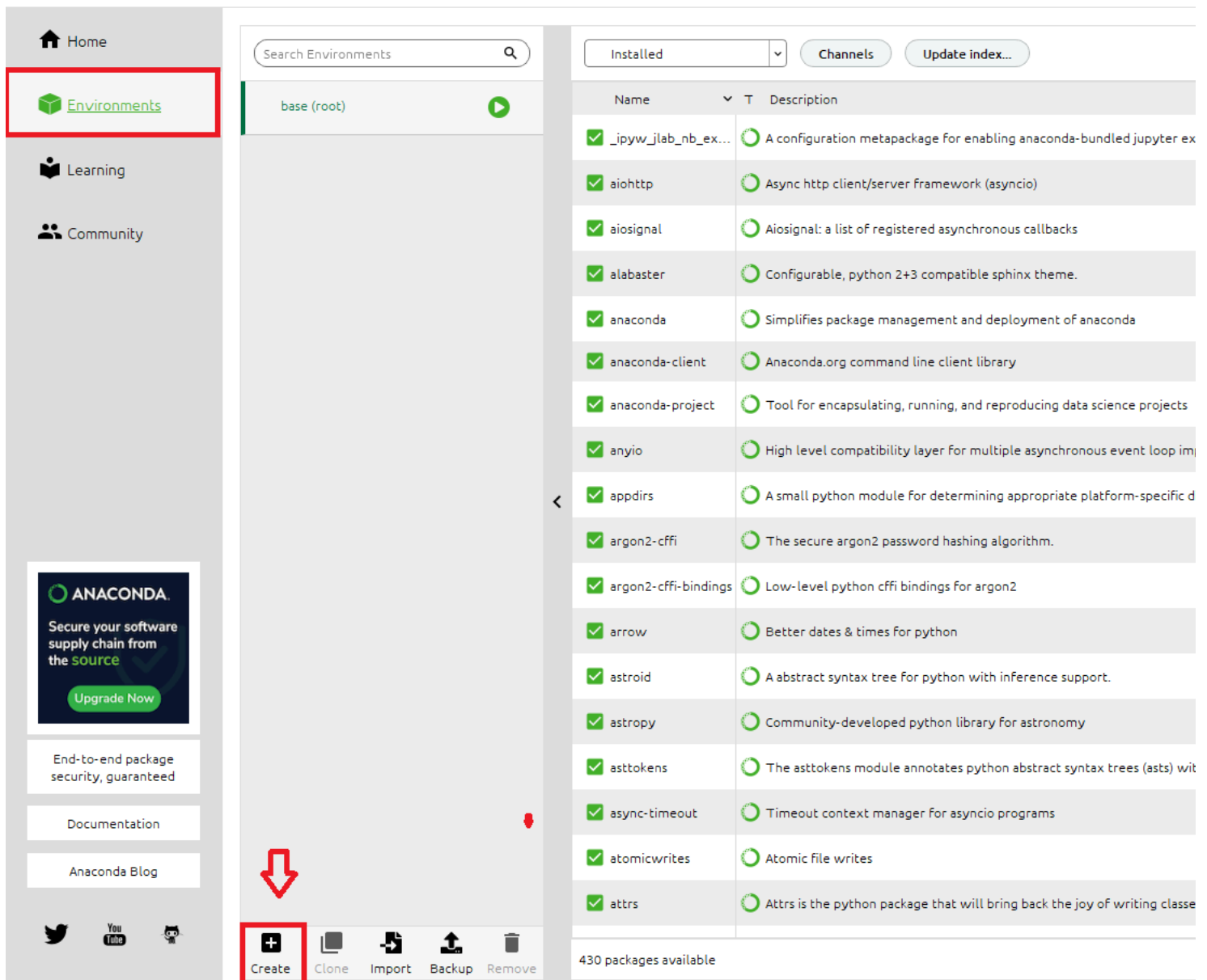
Streamlined co
development c
task runnin

RStudio

1.1.456

A set of integrated tools designed to help
you be more productive with R. Includes R
essentials and notebooks.

Step 2: Create an environment using Anaconda Navigator. Go to the **Environments** tab and click **Create** (at the bottom menu as highlighted below) to create an icon on the Anaconda environment.



The screenshot shows the Anaconda Navigator application interface. On the left sidebar, the 'Environments' tab is highlighted with a red box. The main panel displays a list of installed packages, including _ipyw_jlab_nb_ex..., aiohttp, aiosignal, alabaster, anaconda, anaconda-client, anaconda-project, anyio, appdirs, argon2-cffi, argon2-cffi-bindings, arrow, astroid, astropy, asttokens, async-timeout, atomicwrites, and attrs. The 'Create' button in the bottom toolbar is highlighted with a red box and a red arrow pointing to it.

Note: All the macOS users, select Update index and all your packages will be updated.

Note: It is always helpful to create a separate environment because different projects require different packages.

Step 3: Give a name for your environment, select the suitable version and language and click Create.

☒	_r-mutex	
☒	argon2-cffi	The secure argon2 password hashing algorithm.
☒	argon2-cffi-bindings	Low-level python cffi bindings for argon2
☒	asttokens	The asttokens module annotates python abstract syntax trees (asts) with the positions of tokens and text in the source code that generated them.
☒	attrs	Attr
☒	backcall	Spe
☒	beautifulsoup4	Py
☒	bleach	Eas
☒	ca-certificates	Cer
☒	certifi	Py
☒	cffi	Foreign function interface for python calling c code.
☒	colorama	Cross-platform colored terminal text
☒	debugpy	An implementation of the debug adapter protocol for python
☒	decorator	Better living through python with decorators.
☒	defusedxml	XML bomb protection for python stdlib modules

Create new environment

Name: RP_Env

Location: C:\Users\Skillup 116\anaconda3\envs\RP_Env1

Packages: ☒ Python 3.9.12 ☐ R 3.6.1

Cancel

Create

Note: The macOS users must uncheck Python and then create the environment.

Step 4: Once you create an Anaconda environment, go back to the **Home Page** and **Launch Jupyter** and create a **Python Notebook** (make sure to select the right environment).

Note: The macOS users need to restart their Anaconda prompt first and then launch their Jupyter Notebook.

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Applications on **RP_Env** Channels

DataLore
Online Data Analysis Tool with smart coding assistance by JetBrains. Edit and run your Python notebooks in the cloud and share them with your team.
[Launch](#)

IBM Watson Studio Cloud
IBM Watson Studio Cloud provides you the tools to analyze and visualize data, to cleanse and shape data, to create and train machine learning models. Prepare data and build models, using open source data science tools or visual modeling.
[Launch](#)

jupyter Notebook 6.4.11
Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis.
[Launch](#)

PyCharm
A Full-fledged IDE for Scientific and Web Development. Supports H...
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Glueviz 1.0.0
Multidimensional data visualization across files. Explore relationships within and among related datasets.
[Install](#)

JupyterLab 3.3.2
An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture.
[Install](#)

Orange 3 3.26.0
Component based data mining framework. Data visualization and data analysis for novice and expert. Interactive workflows with a large toolbox.
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PowerShell
Run a Powershell script in current environment.
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Spyder 5.1.5
Scientific Python Development Environment. Powerful Python IDE with advanced editing, interactive testing, and more.
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Step 5: This opens **Jupyter Notebook** in the default browser, and now you can select the **kernel** and create a **Notebook**.

jupyter

Files Running Clusters

Select items to perform actions on them.

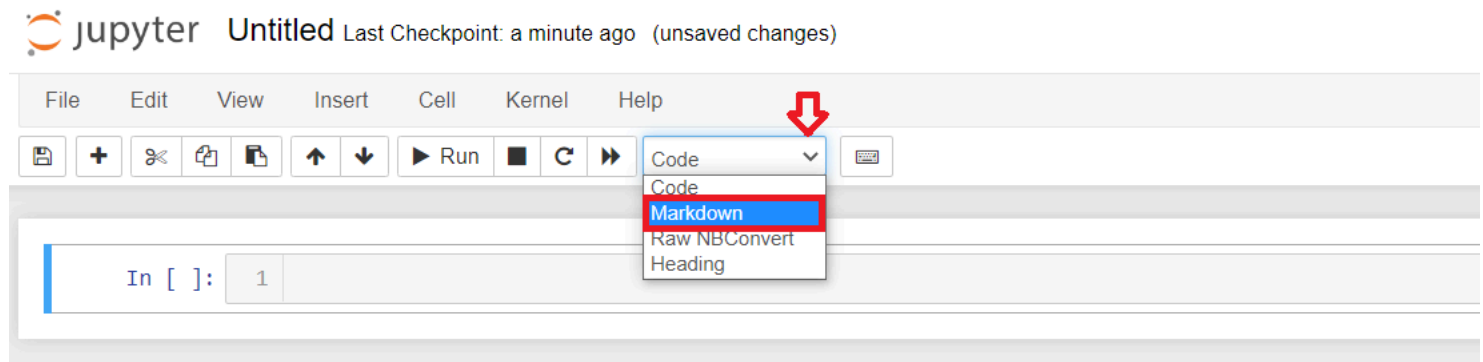
☐ 0 /

- ☐ 3D Objects
- ☐ anaconda3
- ☐ Contacts
- ☐ Desktop
- ☐ Documents
- ☐ Downloads
- ☐ Favorites

Exercise 3: Create and execute Python Jupyter Notebook

Step 1: Create markdown cells and add text

In your notebook, **click any code cell**, and in the drop-down menu, change the cell type from Code to Markdown. You will notice that you cannot create Markdown cells without first creating and converting them from Code to Markdown.



In the Markdown cell, write some text like **My First Program**.

To render the Markdown text, make sure the cell is selected (by clicking within it), and press **Play** in the menu or **Shift+Enter**.

```
# My First Program
```

Your Markdown cell should now be rendered!

► Output

***Note:** To edit your Markdown cell, double-click anywhere within the cell. Note you can use the keyboard shortcut: `[m]` - Convert Cell to Markdown.*

Step 2: Create new cells.

- In your Jupyter Notebook, click any of the existing cells to select the cell.
- Click **Insert Cell Above** or **Insert Cell Below** to insert the cell from the Insert menu.

► Output

***Note:** You can use the keyboard shortcuts: `[a]` - Insert a Cell Above; `[b]` - Insert a Cell Below.*

Step 3: Write and execute code.

- In your new empty notebook, click within the gray code cell and write some code, like.

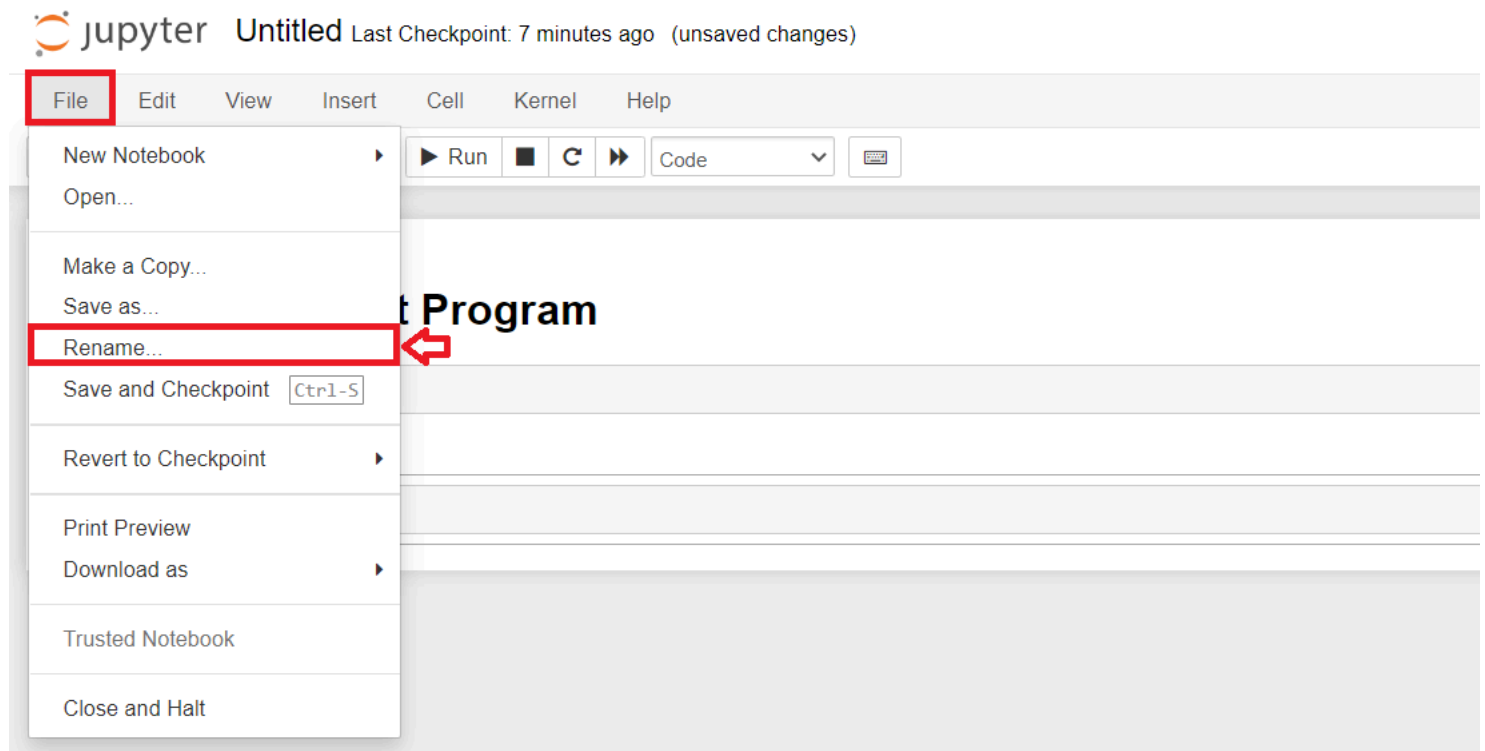
```
1+1
```

- Execute the code by clicking the **Play** button in the menu above the notebook or pressing **Shift+Enter** on your notebook.
- You should see the output 2.

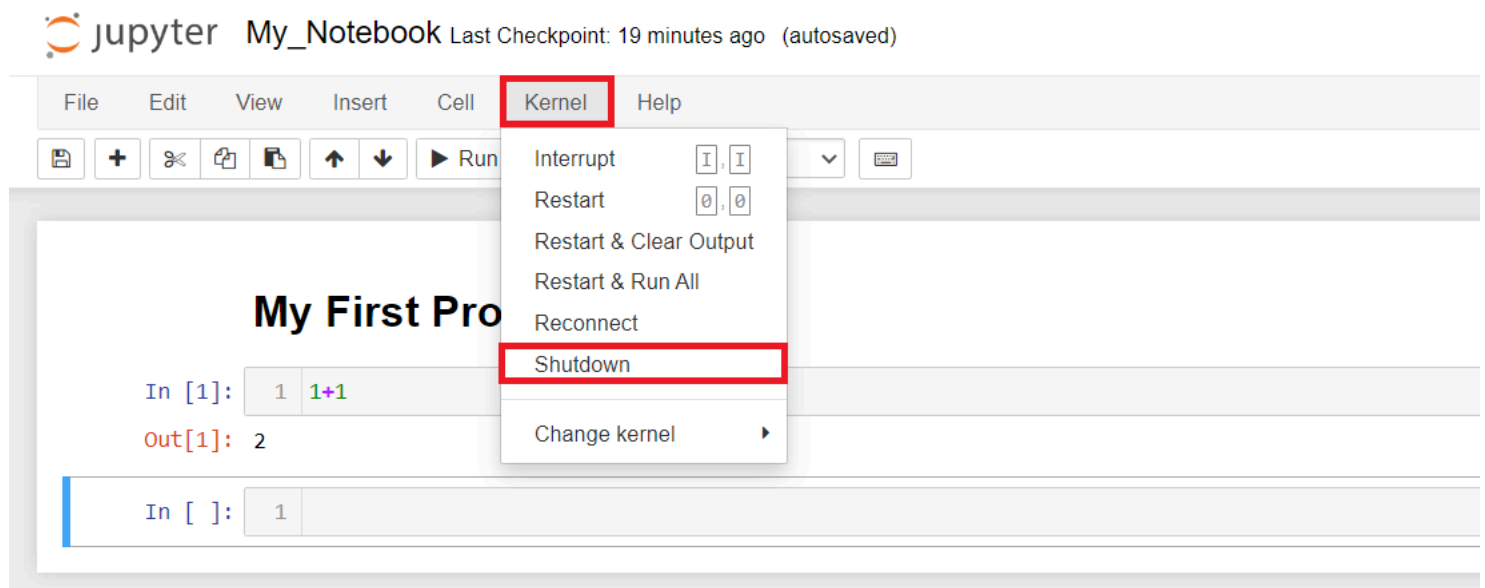
► Output

4. Rename, Shutdown kernel, and Save your Notebook

Step 1: Click **Rename** from the **File** menu to rename your notebook like *My_Notebook.ipynb*.

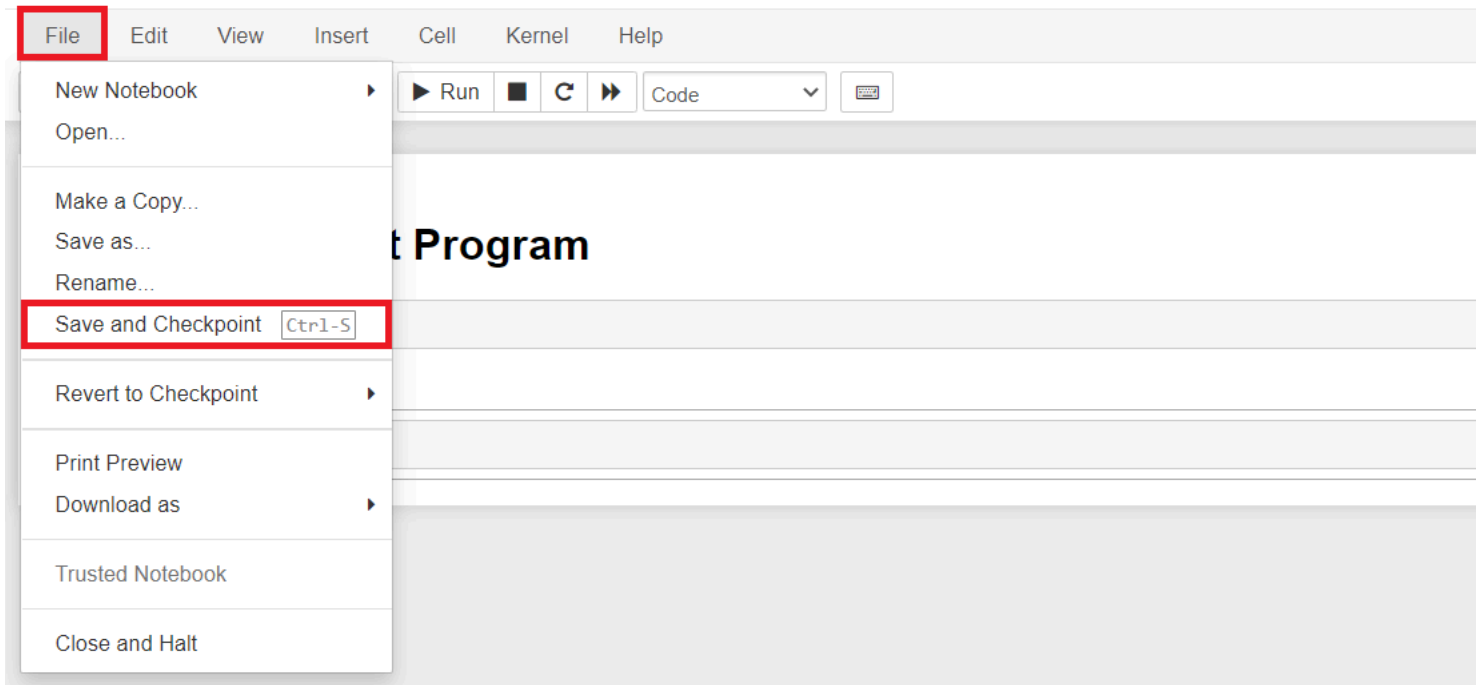


Step 2: To shut down the kernel, click **Shutdown** from the **Kernel** menu.



Step 3: Click **Save Notebook** or **Save Notebook as** to save the notebook from the **File** menu.

 **jupyter** My_Notebook Last Checkpoint: 4 minutes ago (autosaved)



5. Open the recently created notebook.

Step 1: Open **Anaconda Navigator** from the Windows **Start** menu and **launch Jupyter**.

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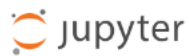
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Step 2: Go to the **directory** where you **saved** your file and **click** to open it.



☐		IBMDeveloperSkillsNetwork-RP0321EN-SkillsNetwork
☐		lax_to_jfk
☐		Links
☐		Maps_with_R
☐		Music
☐		OneDrive
☐		OneDrive - Flexible Road LLC
☐		PycharmProjects
☐		Saved Games
☐		seaborn-data
☐		Searches
☐		Tracing
☐		Videos
☐		Week3
☐		With_R
☐		My_Notebook.ipynb
☐		-1.14-windows.xml
☐		BullseyeCoverageError.txt

Practice Exercise

Let us try executing simple math operations

Problem 1: Find the minimum and maximum values.

```
x = min(5, 10, 25)
y = max(5, 10, 25)
print(x)
print(y)
```

► Output

Problem 2: Find the value of 4 to the power 3.

```
x = pow(4, 3)
print(x)
```

► Output

Exercise 4: Create and execute R Jupyter Notebook

Select the kernel and create a Notebook.



Files

Running

Clusters

Select items to perform actions on them.

☐	0		☐	/
☐	☐	3D Objects		
☐	☐	anaconda3		
☐	☐	Contacts		
☐	☐	Desktop		
☐	☐	Documents		
☐	☐	Downloads		

Problem 1: Find the Multiplication of 2 numbers.

```
2 * 3 # Multiplication
```

► Output

Problem 2: Find the Subtraction of 2 numbers.

```
4 - 1 # Subtraction
```

► Output

Problem 3: Add 2 to the given number.

```
a <- 1 # Assigning 1 to the variable called "a"  
a + 2 # Adding 2
```

► Output

Problem 4: Create a data frame

```
df = data.frame(Emp_Name = c("Jai", "David", "Michael"),  
                Job_role = c("Manager", "Team Lead", "Developer" )  
                )  
print(df)
```

► Output

Congratulations! You have learned how to download and install Anaconda on your local machine and create a new environment. You have also created a Jupyter Notebook and saved it.

Author(s)

[D.M.Naidu](#)



Skills Network